Key features of parabolas

1

a Downwards **b** Upwards

2 Upward

3 No, because the entire parabola will be above the x-axis. Also, there is no solution for the equation $x^2 + 6 = 0$.

4

a No **b** Yes **c** No **d** Yes

5

a $x = 0, 4$ **b** $y = 0$ **c** 4

6

a Concave up **b** $y = 9$ **c** $y = 1$ **d** $x = -2$

e $x < -2$

7

a $(-2, 0)$ **b** Vertex: $(-2, 0)$

c

i False **ii** True **iii** True **iv** True

8

a 2 **b** 1

9 Concave up

10

a $(-4, 2)$ **b** 2

11 $x = 3$

12

a No **b** No **c** Yes

13 Quadratic function A

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15 $y = 25$

16

a $x = 14, 6$ **b** $x = 10$ **c** Concave down **d** $y = 16$

17 The x-coordinate of the vertex of the parabola occurs at $x = \dfrac{-b}{2a}$. The y-coordinate of the vertex is found by substituting $x = \dfrac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x.

18 $x = -3, 2$

19

a False **b** True

20 $x = \dfrac{-3 \pm \sqrt{93}}{6}$

Graphing quadratics

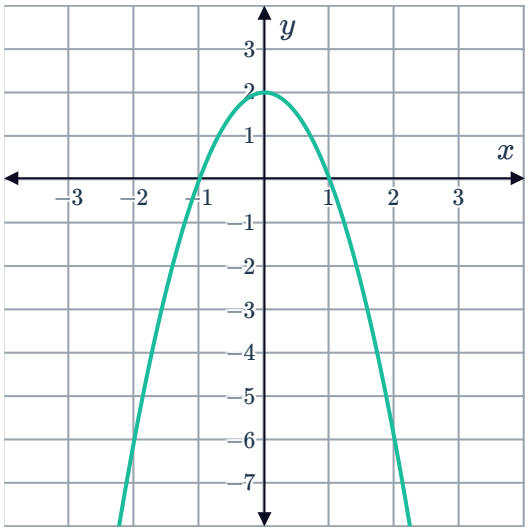
21

a Concave down

b More steep

c $(0, 2)$

d



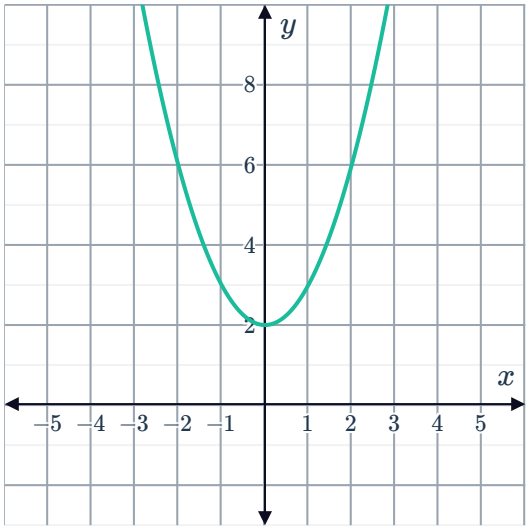
22

a $y = x^2 + 5$ **b** $x \approx 2.3, -2.3$

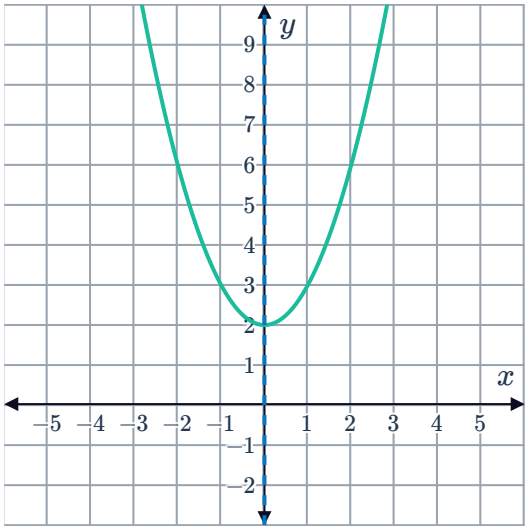
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23

a



b



c $(0, 2)$

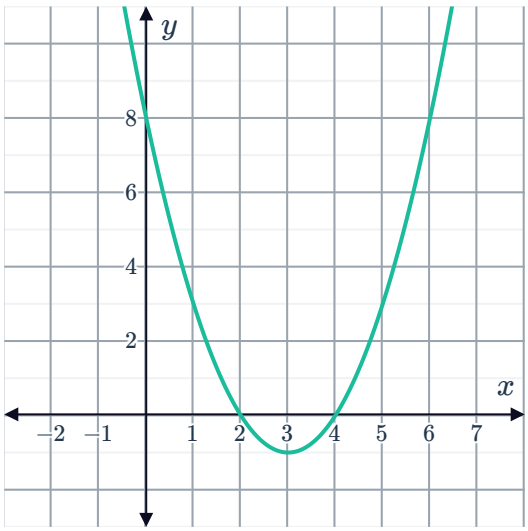
24

a $x = 4, 2$

b $y = 8$

c $(3, -1)$

d



e $x = 4, 2$

25

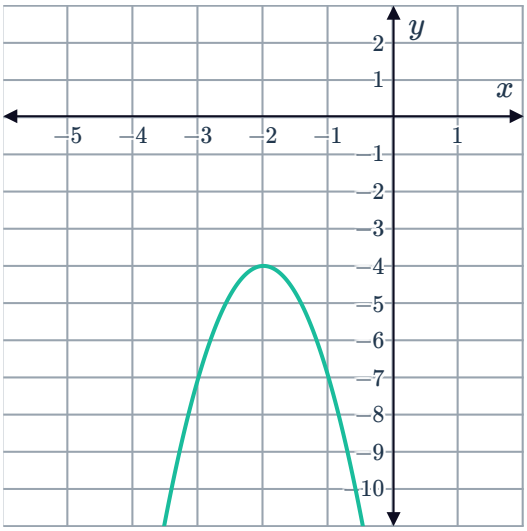
a $(-2, -4)$

b $x = -2$

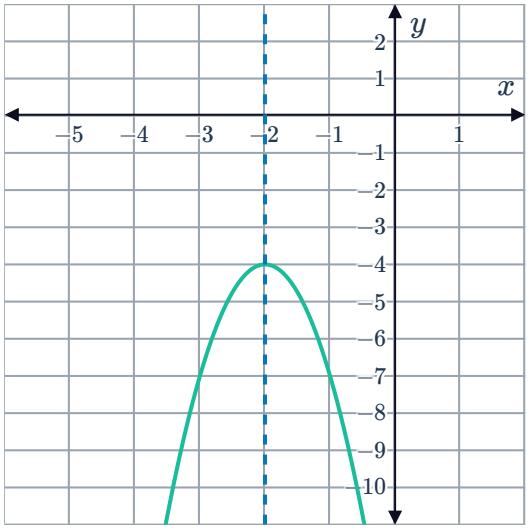
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c $y = -7$

d

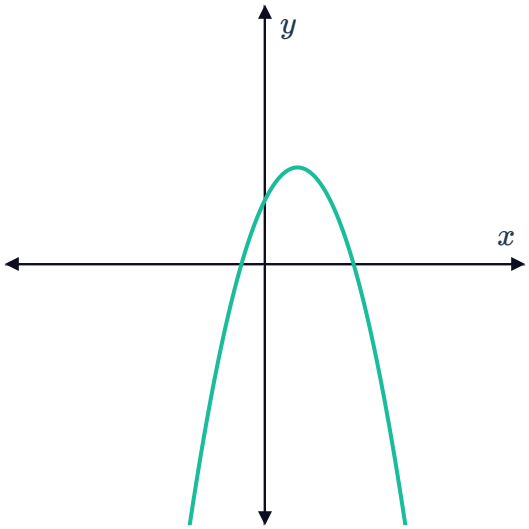


e

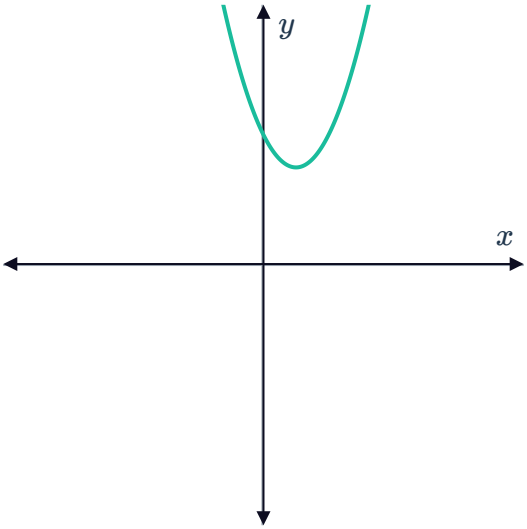


26

a

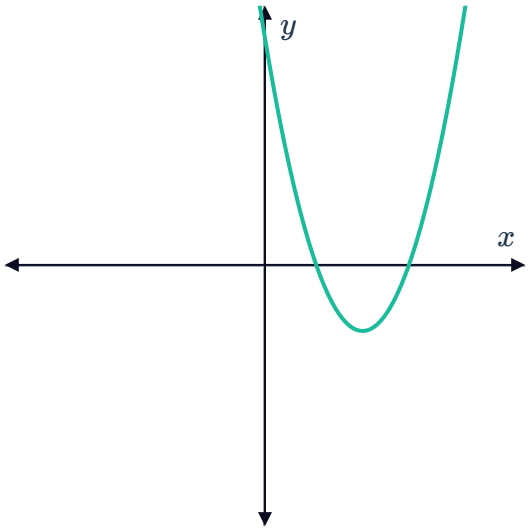


b

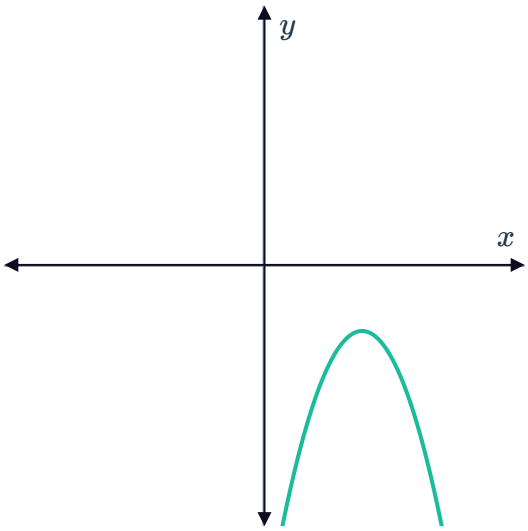


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c



d



27

a 8

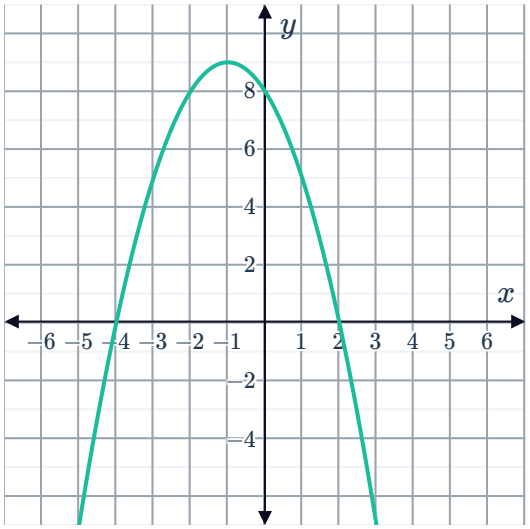
b $x = 2, -4$

c

x	-5	-3	-1	1	3
y	-7	5	9	5	-7

d $(-1, 9)$

e



28

a $y = 3$

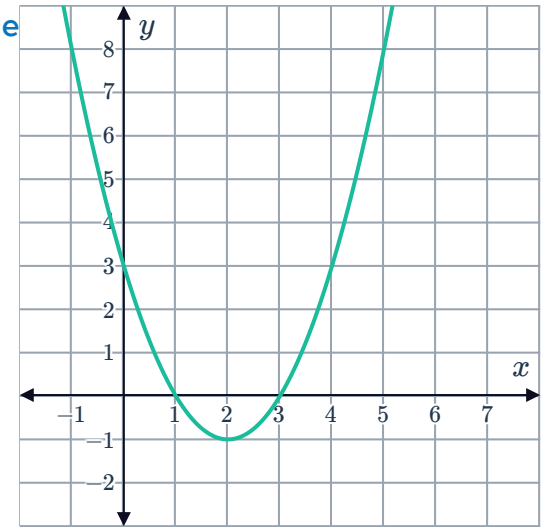
b $x = 3, 1$

c $x = 2$

d $(2, -1)$

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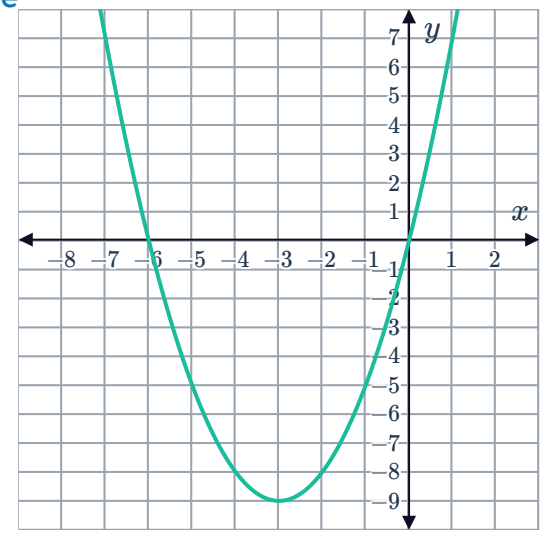
e



29

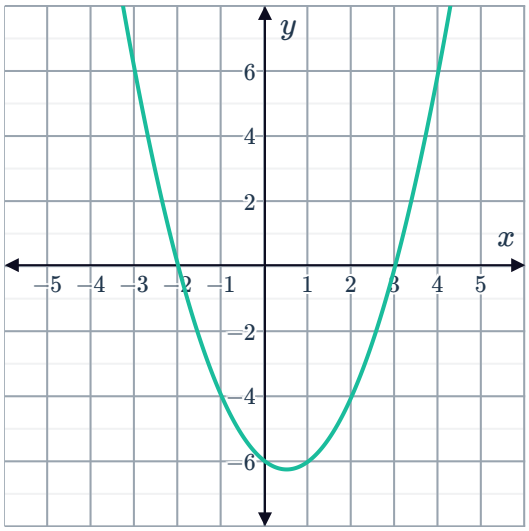
a $y = 0$ **b** $x = 0, -6$ **c** $x = -3$ **d** $(-3, -9)$

e

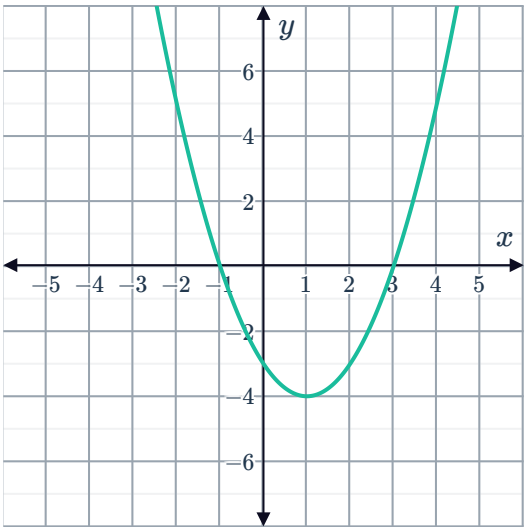


30

a



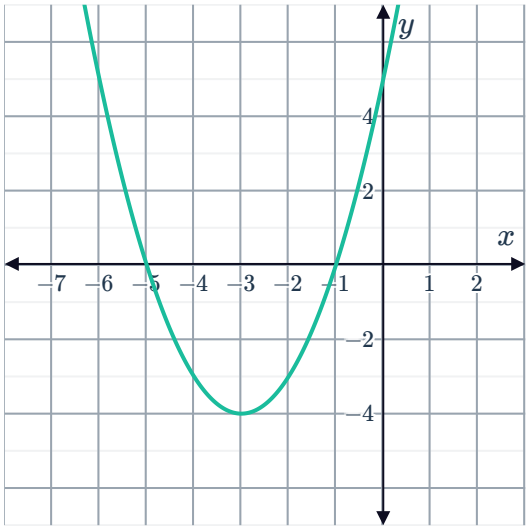
b



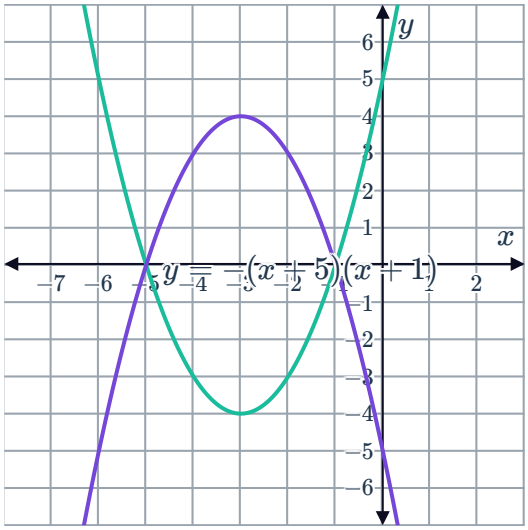
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31

a



b



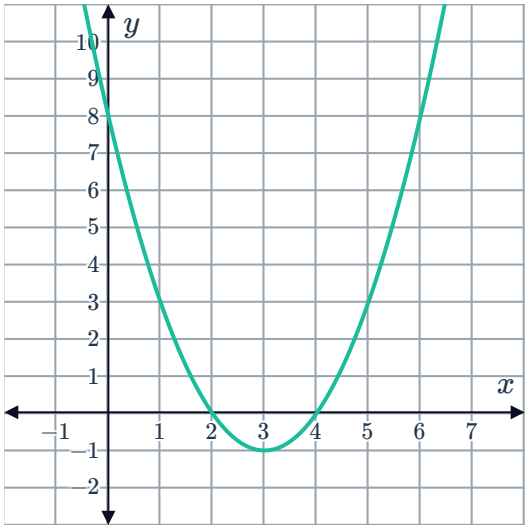
32

a $(x-2)(x-4)$

b $x = 2, 4$

c $(3, -1)$

d



33

a $x = -4, 3$

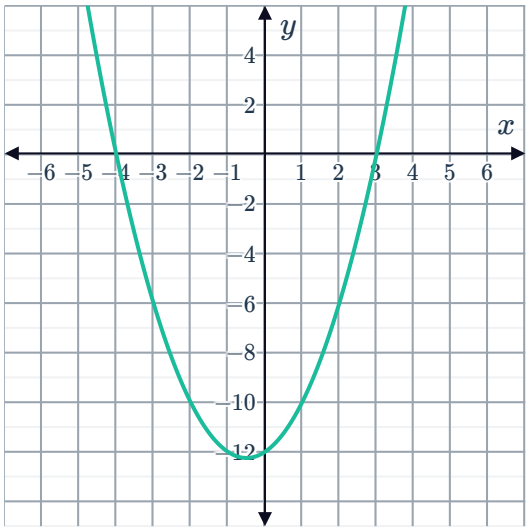
b $y = -12$

c $x = -\dfrac{1}{2}$

d $\left(-\dfrac{1}{2}, -\dfrac{49}{4}\right)$

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e



34

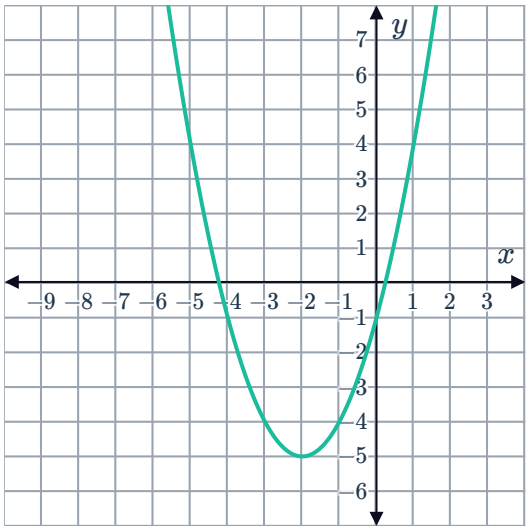
a $y = (x+2)^2 - 5$

b $y = -1$

c $(-2, -5)$

d Concave up

e



f $x \approx 0.2, -4.2$

35

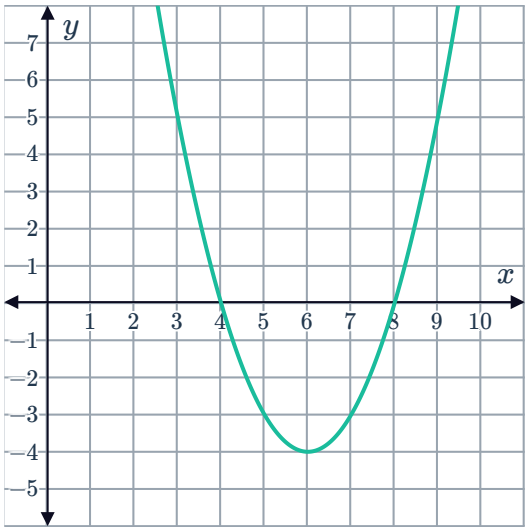
a $x = 8, 4$

b $y = (x-6)^2 - 4$

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c $(6, -4)$

d

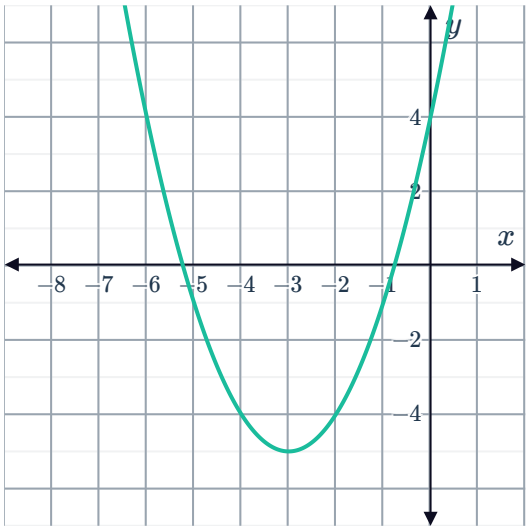


36

a $x = -3$

b -5

c

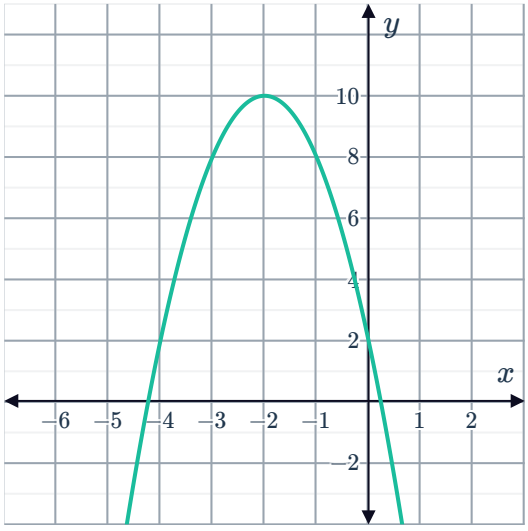


d $x \approx -0.8, -5.2$

37

a $(-2, 10)$

b



c $x \approx 0.3, -4.3$

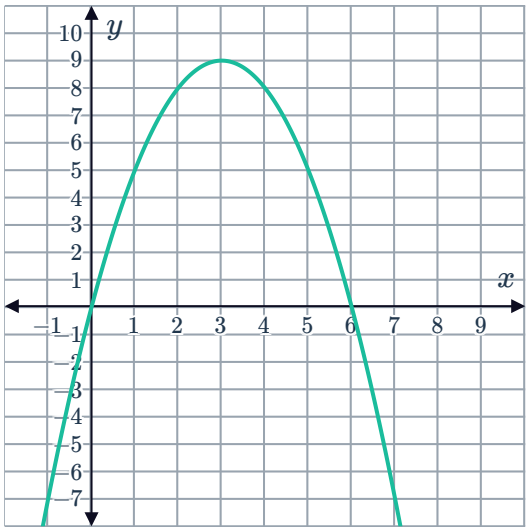
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38

a $x = 0, 6$

b $(3, 9)$

c



39

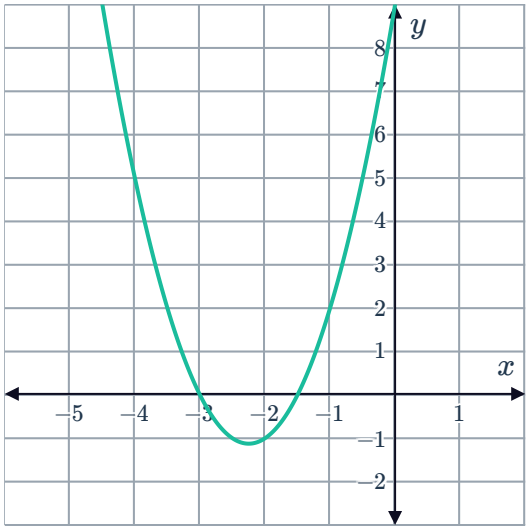
a $x = -\frac{3}{2}, -3$

b $y = 9$

c $x = -\frac{9}{4}$

d $y = -\frac{9}{8}$

e



Graphing quadratics using technology

40

a

i $(8, 7)$ **ii** $(0, 263)$

b

i $(-6, 4)$ **ii** $(0, -140)$

41

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a $(0, -12)$ **b** No **c** $x < 0$

42

a $x = 2, 6$ **b** $y = -24$ **c** $x = 4$ **d** $y = 8$

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